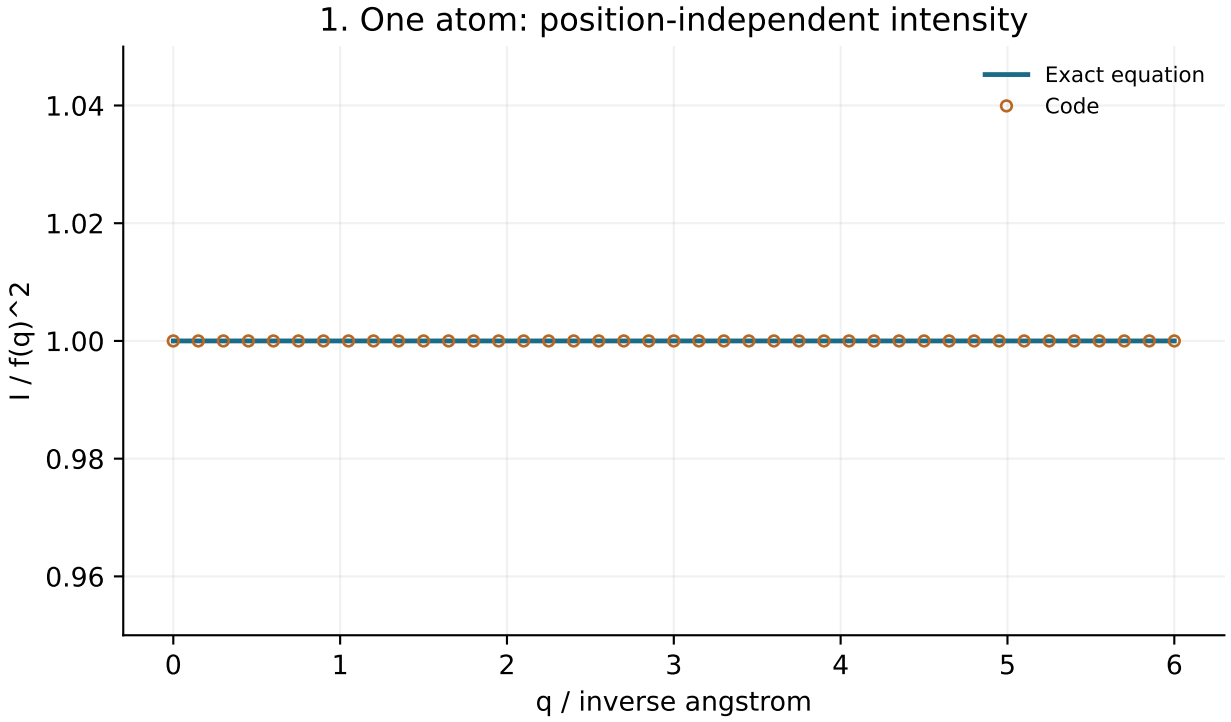
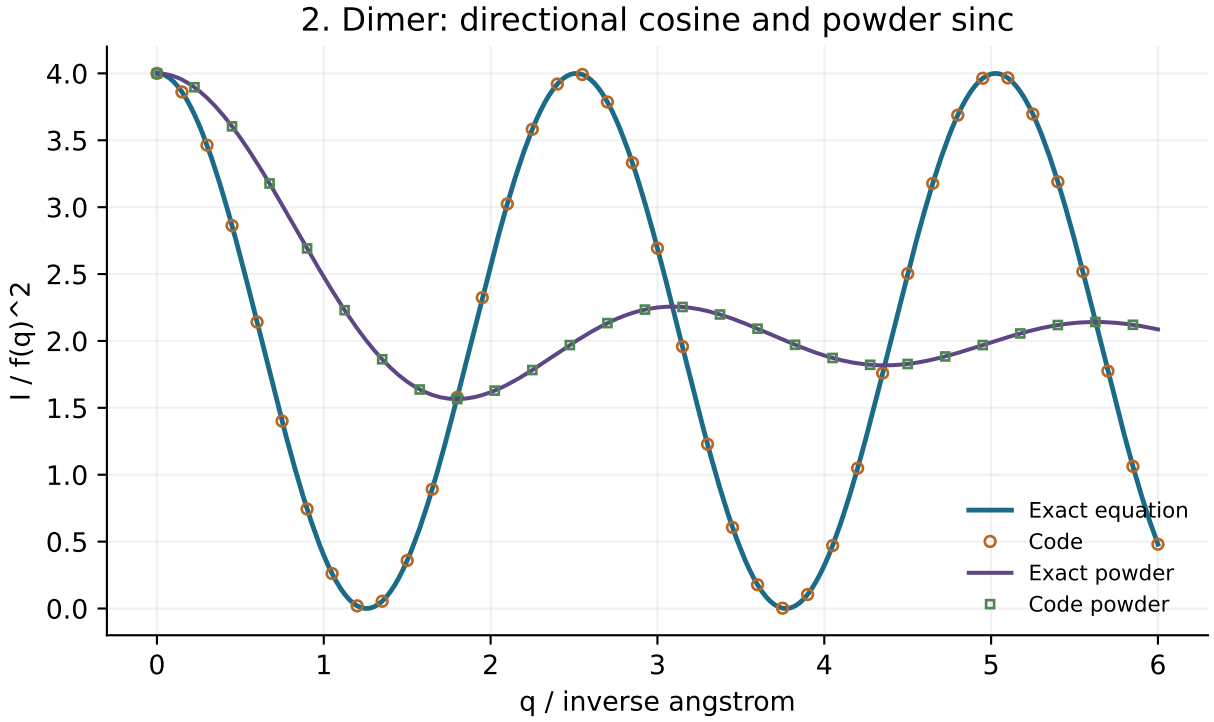
Known physical solutions: equations versus computed scattering

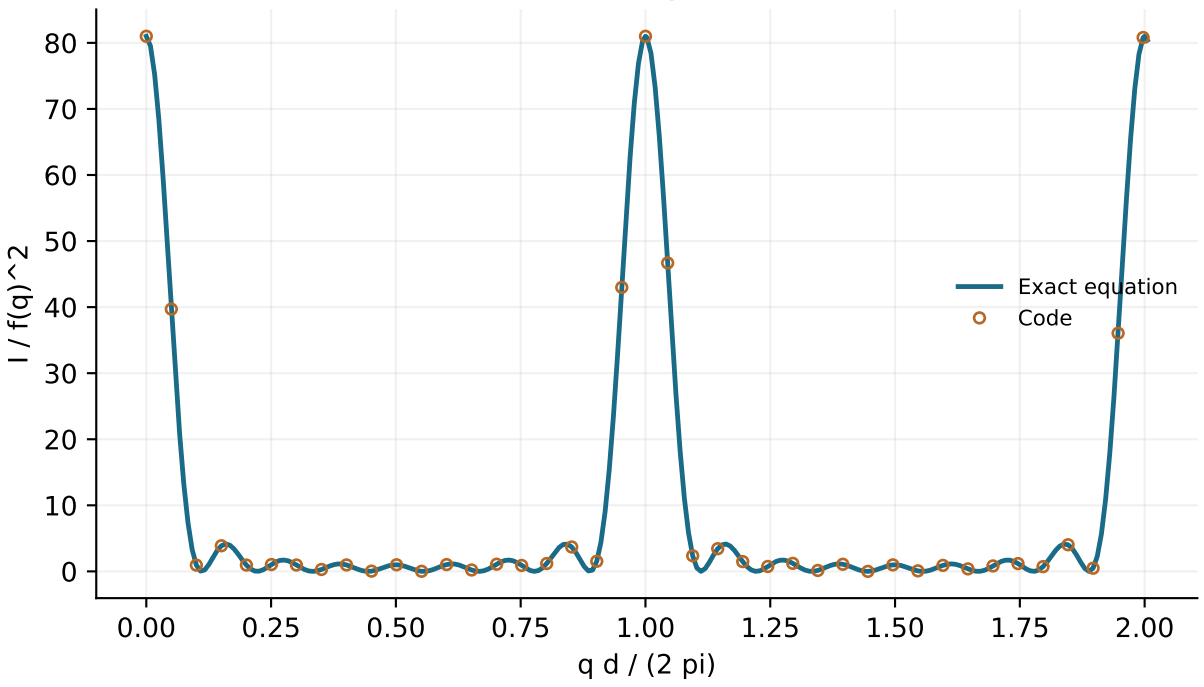
1. One atom: position-independent intensity



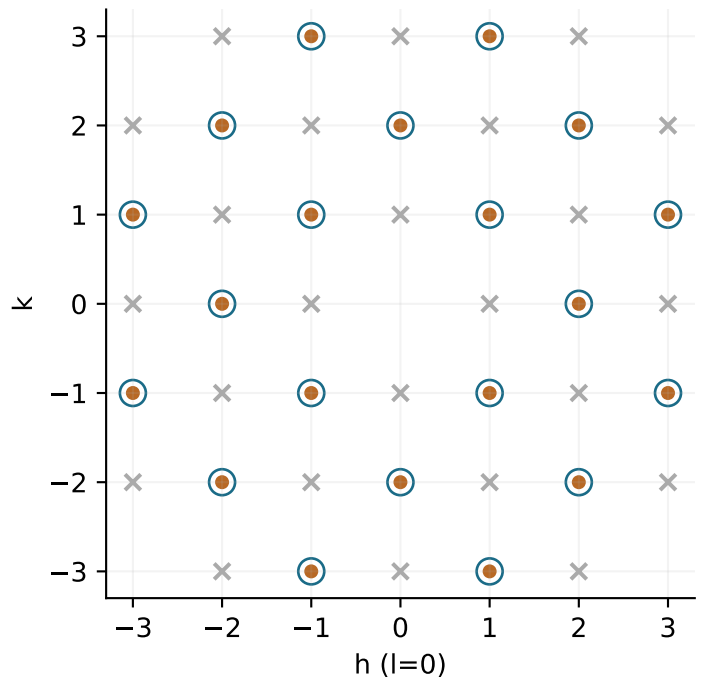
2. Dimer: directional cosine and powder sinc



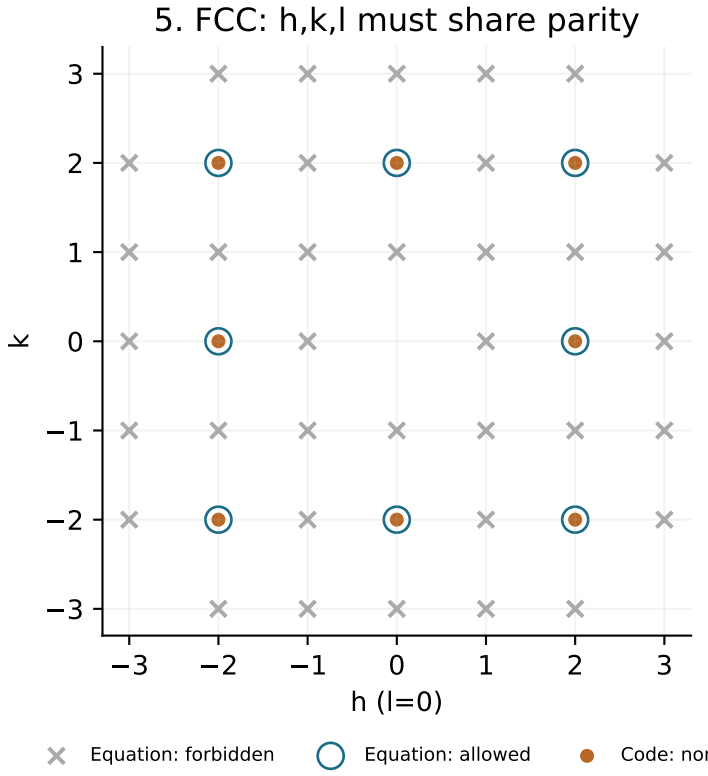
3. Nine-atom chain: fringes and N^2 maxima



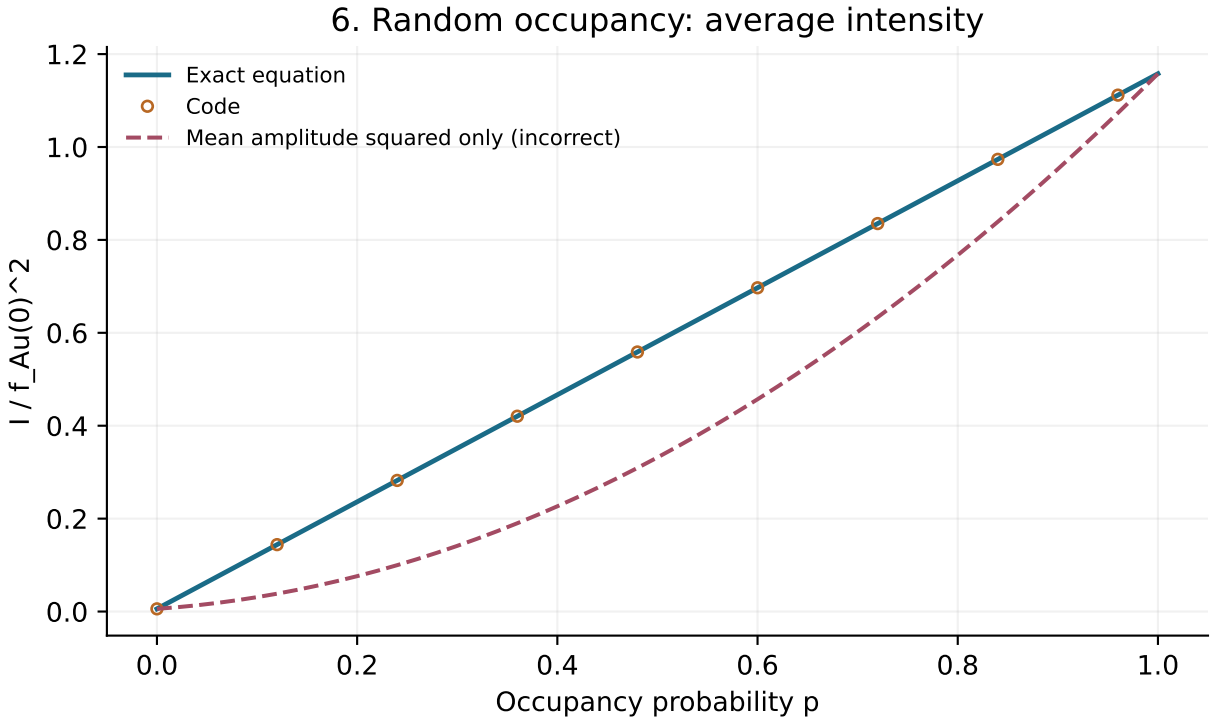
4. BCC: h+k+l must be even



5. FCC: h,k,l must share parity



6. Random occupancy: average intensity



7. Gaussian disorder: coherent to diffuse

8. Uniform-ball guest: sampled versus exact